$2x + 78.4 - 5x = 32$ l

$78.4 - 3x = 32$

$3x = 46.4$ l

$x = 15.47$ l

Given mixture contains 15.47 lit of methane and 0.213 lit of propane, hence

$$\Delta H_C = \left[\left(\frac{\Delta H_C\ (CH_4)}{22.4 \text{ lit}}\right) x \text{ lit}\right] + \left[\left(\frac{\Delta H_C\ (C_3H_8)}{22.4 \text{ lit}}\right)(15.68 - x) \text{ lit}\right]$$

$$\Delta H_C = \left[\left(\frac{-890 \text{ kJ mol}^{-1}}{22.4 \text{ lit}}\right) 15.47 \quad \text{lit}\right] + \left[\left(\frac{-2220}{22.4 \text{ lit}}\right) 0.21 \quad \text{lit}\right]$$

$$\Delta H_C = \left[-611.87 \quad \text{kJ mol}^{-1}\right] + \left[-20.81 \quad \text{kJ mol}^{-1}\right]$$

$$\Delta H_C = -632.68 \text{ KJ mol}^{-1}$$

19.

$4E_{C-H} = 360 \text{ kJ mol}^{-1}$

$E_{C-H} = 90 \text{ kJ mol}^{-1}$

$E_{C-C} + 6\ E_{C-H} = 620 \text{ kJ mol}^{-1}$

$E_{C-C} + 6 \times 90 = 620 \text{ kJ mol}^{-1}$

$E_{C-C} + 540 = 620 \text{ kJ mol}^{-1}$

$E_{C-C} = 80 \text{ kJ mol}^{-1}$

22. During compression, energy of the system increases, in isothermal condition, to maintain temperature constant, heat is liberated from the system. Hence q is negative.

During compression entropy decreases.

During compression work is done on the system, hence w is positive

23.

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$T_b = \frac{\Delta H_V}{\Delta S_V} = \frac{4800 \text{ J mol}^{-1}}{16 \text{ J mol}^{-1} \text{ K}^{-1}} = 300 \text{ K} = 27^0 \text{C}$$

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